

MANUFACTURING DATA JOURNEY

FROM DATA TO INSIGHTS

Transforming Industrial Data into
Actionable Insights for Smarter
Manufacturing Decisions



PRODUCTION



QUALITY



MAINTENANCE



INVENTORY



COST



Alaa Hamed
Industrial Data Analysis

SQL Server | Power BI | Python | Manufacturing Analytics



Project Overview

End-to-end Industrial Data Analytics for a chocolate manufacturing environment

DATA PLATFORM

SQL Server Data Warehouse

ANALYTICAL MODEL

Star Schema + ETL

ANALYTICS LAYER

Power BI Dashboards

- Transform raw manufacturing data into reliable analytical information.
- Centralize production, maintenance, quality, inventory, finance and HSE data.
- Provide management and operational visibility through interactive dashboards.
- Create a foundation for future predictive maintenance and machine learning.

Business Scope & Objectives

BUSINESS DEPARTMENTS

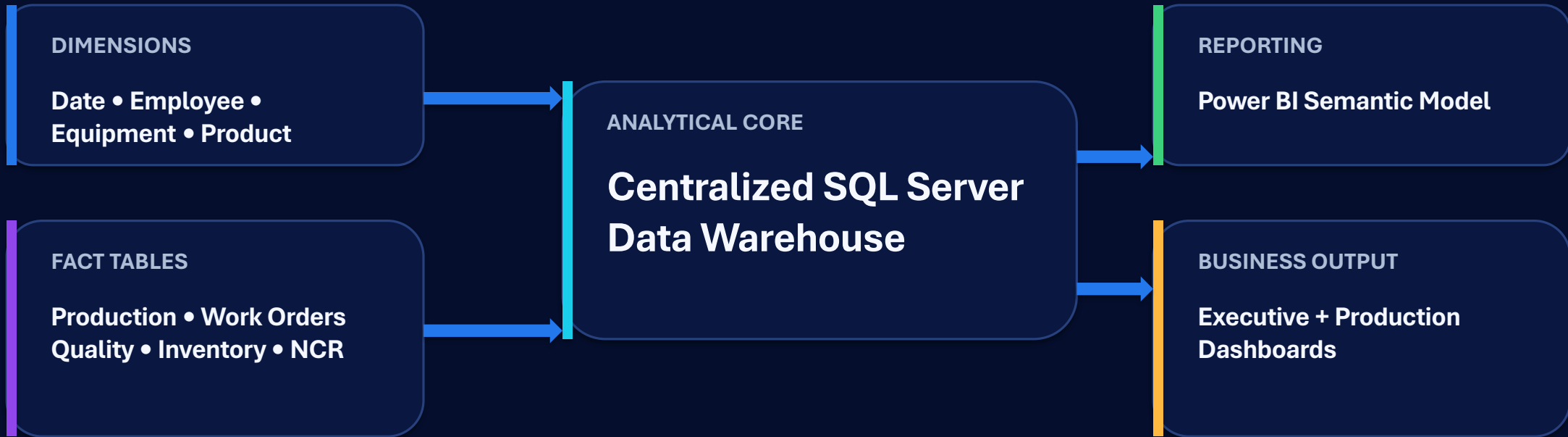
Maintenance
Production
Quality
Warehouse / Inventory
Finance
HSE — Health, Safety & Environment

BUSINESS OBJECTIVES

Build a centralized enterprise data warehouse
Improve data quality and reporting reliability
Monitor manufacturing KPIs
Analyze production performance, costs and quality
Prepare structured data for advanced analytics

Data Warehouse Architecture

Dimensional modeling using a Star Schema



ETL Workflow

Structured flow from raw CSV files to analytical data

Extract

01

Raw CSV files

Stage

02

Load staging tables

Clean

03

Data cleaning

Validate

04

Data validation

Master

05

Load master tables

Warehouse

06

Populate fact/dimension tables

Model

07

Create relationships

Analyze

08

Build Star Schema + Power BI

Power BI — Executive Manufacturing Overview

Management-level view of manufacturing performance



- Total Production Orders
- Total Produced Quantity
- Average Yield %
- Total Manufacturing Cost
- Production Performance — Actual vs Plan
- Manufacturing Cost Variance by Year
- Quality Acceptance % by Year
- Maintenance Work Orders & Downtime

Power BI — Production Performance Dashboard

Operational analysis of production output, yield and product performance

PRODUCTION ACHIEVEMENT

98.40%

PRODUCED QUANTITY

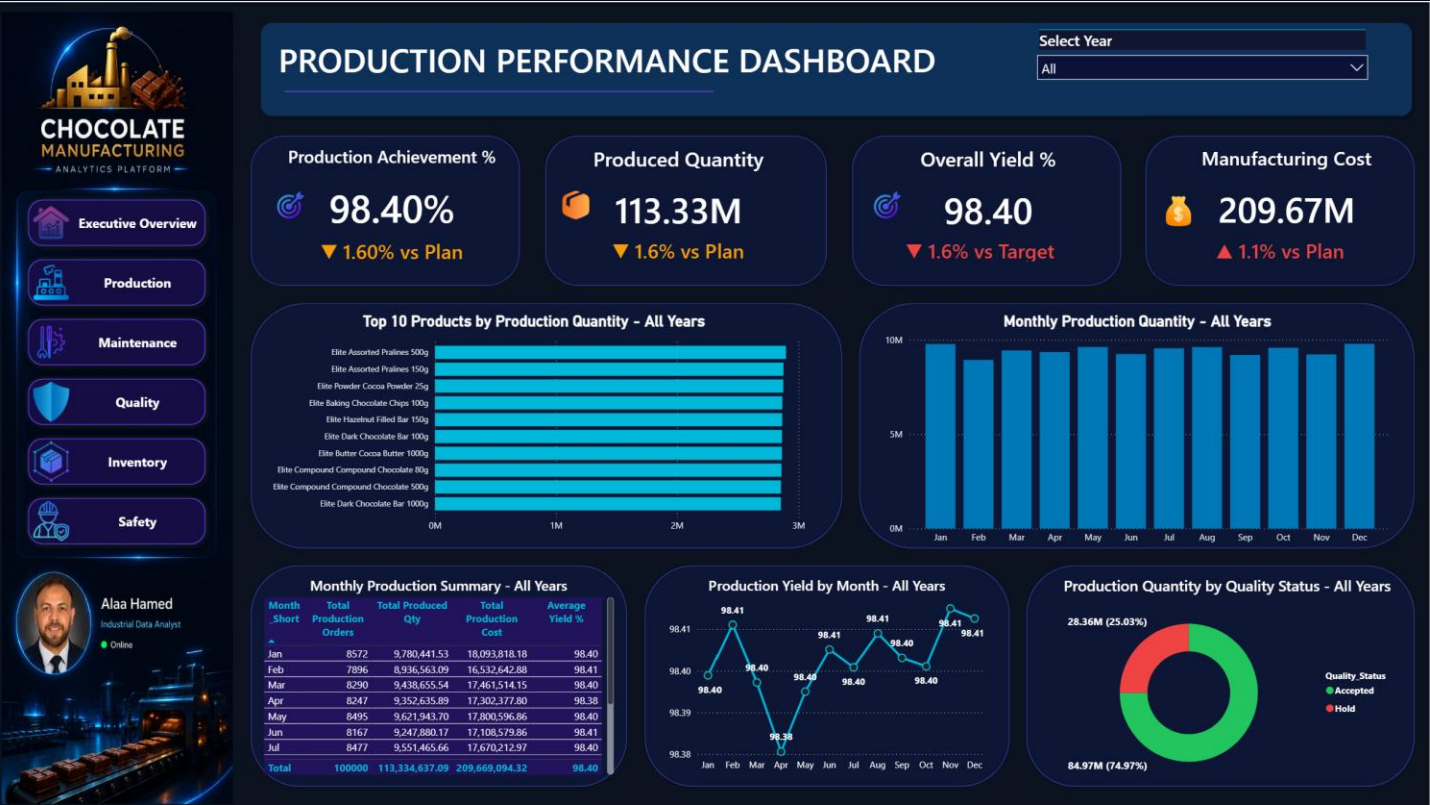
113.33M

OVERALL YIELD

98.40%

MANUFACTURING COST

209.67M



Key Manufacturing Insights

What the analytical layer reveals

PRODUCTION PERFORMANCE

98.40% of plan

Strong overall production performance, with a small gap to plan.

YIELD PERFORMANCE

98.40% • ~1.60% below target

Yield remains high, but the gap points to an efficiency improvement opportunity.

MANUFACTURING COST

209.67M • ~1.14% above plan

Cost performance should be monitored together with production performance.

QUALITY ACCEPTANCE

~75% range

Quality acceptance improved after 2021 and remained around the mid-70% range.

- The dashboard links operational KPIs to business targets.
- Trend views help identify where deeper root-cause analysis is needed.
- The same data foundation can support predictive analytics.

Business Impact & Next Steps

TODAY

Centralized manufacturing data

Reliable KPI reporting

Interactive Power BI visibility

Production / Quality / Cost monitoring

NEXT

Deeper root-cause analysis

Cross-department analytics

Automated refresh & reporting

Stronger KPI governance

FUTURE

Predictive Maintenance

Machine Learning

Failure-risk forecasting

Proactive maintenance decisions

Portfolio & Project Access

The presentation is supported by the complete project repository

GITHUB REPOSITORY

Industrial_Data_Warehouse_Chocolate_Manufacturing

https://github.com/alaahamed780-lgtm/Industrial_Data_Warehouse_Chocolate_Manufacturing

PORTFOLIO STRUCTURE

**Datasets • SQL Scripts • Database Diagram
Documentation • Power BI • Screenshots • Presentation**

CORE TECHNOLOGIES

SQL Server 2022
T-SQL
Excel
Power BI
Star Schema
ETL
Data Validation
Manufacturing Analytics
Git & GitHub